THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM THREE
041 BASIC MATHEMATICS
MARKING SCHEME AUGUST 2026

SECTION A (60 Marks)

1. (a) The LCM of 15, 18, and 21

2	15	18	21
*3	15	9	21
3	5	3	7
5	5	1	7
7	1	1	7
	1	1	1

mark

The GCF= 3.....$00\frac{1}{2}$

.....01 mark

The LCM = $2 \times 3 \times 3 \times 5 \times 7 = 630$ .....$00\frac{1}{2}$ mark

Sum= $LCM + GCF = 630 + 3 = 633$ .....01 mark

(b) *let* $x = 2.\overline{45}$

$x = 2.\overline{45}$ .....equation 1

$100x = 2.\overline{4545} \times 100$

$100x - x = 245.\overline{45}\ \ 99x = 245.\overline{45}$ .....Equation2.....$00\frac{1}{2}$ mark

Subtract equation 1 from 2

$$\begin{array}{l} 100x = 245.\overline{45} \\ -\ \ x = 2.\overline{45} \end{array}$$

$99x = 243$

$\frac{99x}{99} = \frac{243}{99}$

$x = \frac{243}{99}$ .....01 mark

$\frac{y}{15} = \frac{243}{99}$

$y = \frac{15 \times 243}{99}$

$= \frac{1215}{11}$.....$00\frac{1}{2}$ mark

$y = 110\frac{5}{11}$ .....01 mark

2. (a) $\frac{1+tan60^\circ}{1-tan60^\circ} = \frac{1+\frac{\sqrt{3}}{3}}{1-\frac{\sqrt{3}}{3}} = \frac{3+\sqrt{3}}{3-\sqrt{3}}$.....01 mark

$= \frac{3+\sqrt{3}}{3-\sqrt{3}} \times \frac{3+\sqrt{3}}{3+\sqrt{3}}$

$= \frac{9+6\sqrt{3}+3}{9-3}$ ……01 mark

$= \frac{12+6\sqrt{3}}{6}$

$= 2 + \sqrt{3}$ ……01 mark

(b) $\log x = \frac{\log(x^2-4x+8)}{\log 100}$

$\log x = \frac{\log(x^2-4x+8)}{2}$

$2\log x = \log(x^2 - 4x + 8)$

$x^2 = x^2 - 4x + 8$ ……01 mark

$4x = 8$

$\frac{4x}{4} = \frac{8}{4}$

$x = 2$……01 mark

3. Let the number of villagers who own TV be T

Let the number of villagers who own car be C

$n(\mu) = 100, n(T) = 75, n(C) = 45, n(T \cap C) = 35$

$n(T \cup C) = n(T) + n(C) - n(T \cap C)$

$= 75 + 45 - 35$……$00\frac{1}{2}$ mark

$= 120 - 35$

$= 85$……01 mark

$n(\mu) - n(T \cup C) = n(T \cap C)'$

$100 - 85 = n(T \cap C)'$……$00\frac{1}{2}$ mark

$15 = n(T \cap C)'$ ……01 mark

4. (a) Finding the Values of m and k

Substitute the given coordinates $(-4,5)$ and $(-2,1)$ into the straight line equation $y = mx + k$:

For $(-4,5)$: $5 = -4m + k$ —— (Equation 1)

For $(-2,1)$: $1 = -2m + k$ —— (Equation 2)……01 mark

Step 2: Solve for m by elimination or substitution

Subtract Equation 2 from Equation 1 to eliminate constant k:

$(5-1) = (-4m - (-2m)) + (k - k) 4 = -2mm = \frac{4}{-2} = -\mathbf{2}$ ……01 mark

Solve for k

Substitute $m = -2$ into Equation 2:

$1 = -2(-2) + k 1 = 4 + k k = 1 - 4 = -\mathbf{3}$ ……01 mark

(b) Finding the Equation of the Line

Rearrange the reference line $4x + y + 6 = 0$ into slope-intercept form ($y = mx + c$):

$y = -4x - 6$

The gradient of this reference line (m_1) is -4.

The y-intercept (c) is $-\mathbf{6}$……01 mark

Determine the new perpendicular gradient (m_2)

$-1{:}m_1 \times m_2 = -1 - 4 \times m_2 = -1m_2 = \frac{1}{4}$ .....01 mark

Substitute the new gradient $m_2 = \frac{1}{4}$ and y-intercept $c = -6$ into the standard slope-intercept form: $y = \frac{1}{4}x - 6$

$x - 4y - 24 = 0$ .....01 mark

5. (a) (i) $\Delta QRP \sim \Delta SUT$

If the measures of two sides of a triangle are proportional to the measures of two corresponding sides of another triangle and the included angles are congruent, then the triangles are similar

i.e $\frac{PQ}{ST} = \frac{OR}{SU}$ and $< Q \cong < S, > O$ .....*01 mark*

$\Delta QRP \sim \Delta SUT$ .....*01 mark*

(ii) $\frac{PQ}{ST} = \frac{QR}{SU}$

$\frac{a}{ax} = \frac{c}{cx}$

$\frac{1}{x} = \frac{1}{x}$ .....*01 mark*

$\overline{PQ} = \overline{ST}$ (given side)

$\overline{QR} = \overline{SU}$ (given side)

PQR $\cong$ TSU (given angle

$\therefore \Delta QRP \sim \Delta SUT$ (SAS Similarity theorem) .....*01 mark*

(b) $\overline{AB} \cong \overline{BC}$ (given side) .....*01 mark*

$\overline{BF} \cong \overline{BE}$ (given side) .....*01 mark*

$\overline{AF} \cong \overline{EC}$ since $\overline{AE} \cong \overline{FC}$ and $\overline{EF}$ is common (given side)

$\therefore \Delta AFB \cong \Delta CEB$ (SSS theorem)

.....*01 mark*

6. (a) Number of trees $= \left(\frac{650\,m}{13\,m} + 1\right) 2$.....01 mark

$= 51(2)$

$= 102$ trees.....01 mark

1 tree $= 3\,500$ shillings

102 trees $= y$

$y = \frac{102\ trees \times 3\,500\ shillings}{1\ tree}$ .....01 mark

$y = 357\,000$ shillings

Therefore he incurred a cost of 357 000 shillings .....01 mark

(b) i. $y \propto \frac{x}{z}, y = \frac{xk}{z}, k = \frac{yz}{x}$

$k = \frac{10 \times 5}{8} = \frac{25}{4}$.....01 mark

$\frac{25}{4} = \frac{yz}{x}, \frac{x}{z} \times \frac{25}{4} = \frac{yz}{x} \times \frac{x}{z}$

$y = \frac{25x}{z}$.....01 mark

ii. $\frac{25}{4} = \frac{y \times 2.5}{6}$, $y = \frac{25 \times 6}{2.5}$

$y = 15$.....01 mark

7. (a) David : Amina $= 7:3$

David : Nuru$= 4:5$

Let the oranges received by Nuru be x

Therefore $x: x + 21 = 4:5$.....$00\frac{1}{2}$ mark

$\frac{x}{x+21} = \frac{4}{5}$ .....$00\frac{1}{2}$ mark

$5x = 4x + 8$

$5x - 4x = 84$

$x = 84$.....$00\frac{1}{2}$ mark

Let the number of oranges received by Nuru be y

$84 : y = 7:3$

$\frac{84}{y} = \frac{7}{3}$ .....01 mark.

$y = \frac{3 \times 84}{7} = 36$

Therefore Amina received 36 oranges.....$00\frac{1}{2}$ mark

(b)Total ratios=6+8+10=24 parts.....$00\frac{1}{2}$ mark

Difference between the largest and smallest $= \left(\frac{10-6}{24}\right) \times 288\ cm$.....01 mark.

$= \frac{4}{24} \times 288\ cm$.....01 mark.

$= 48\ cm$ ....$00\frac{1}{2}$ mark

8. (a)$A_3 = A_1 + 2d, A_2 = A_1 + d$

$A_3 + A_2 = A_1 + 2d + A_1 + d = 19$

$2A_1 + 3d = 19$.....Equation 1..... $00\frac{1}{2}$ mark

$A_5 = A_1 + 4d,\ A_5 - 2(A_2) = 8$

$A_1 + 4d - 2(A_1 + d) = 8$

$A_1 + 4d - 2A_1 - 2d = 8$

$-A_1 + 2d = 8$.....$00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$\begin{cases} 2A_1 + 3d = 19 \\ -A_1 + 2d = 8 \end{cases}$

$2 \left| \begin{cases} 2A_1 + 3d = 19 \\ -A_1 + 2d = 8 \end{cases} \right.$

$+ \left| \begin{cases} 2A_1 + 3d = 19 \\ -2A_1 + 4d = 16 \end{cases} \right.$

$7d = 35,\ \frac{7d}{7} = \frac{35}{7},\ d = 5$.....$00\frac{1}{2}$ mark

Substitute 5 for d in equation 2

$-A_1 + 2(5) = 8, A_1 = 10 - 8 = 2$.....$00\frac{1}{2}$ mark

$S_n = \frac{n}{2}[2A_1 + (n-1)d]$

$= \frac{50}{2}[2 \times 2 + (50-1)5]$

$= 25[4 + 49 \times 5]$.....00$\frac{1}{2}$ mark

$= 25[249]$

$= 6225$.....00$\frac{1}{2}$ mark

(b)$G_1 = \frac{1}{64}, G_2 = \frac{-1}{32}, G_3 = \frac{1}{16}$

$r = \frac{G_2}{G_1} = \frac{-1}{32} \times \frac{64}{1} = -2$.....01 mark

$G_n = G_1 r^{n-1}$

$G_8 = \frac{1}{64}\left[(-2)^{8-1}\right]$.....01 mark

$= \frac{-128}{64}$

$= -2$.....01 mark

9. (a) $a^2 + b^2 = c^2$

$3^2 + 4^2 = y^2$

$9 + 16 = y^2$

$25 = y^2$

$\sqrt{25} = \sqrt{y^2}$

$\pm 5 = y$

$y = 5, y \geq 0$.....01 mark)

$a^2 + b^2 = c^2$

$5^2 + 12^2 = c^2$

$25 + 144 = c^2$

$169 = c^2$

$\sqrt{169} = \sqrt{c^2}$

$\pm 13 = c$

$y = 13\ cm, c \geq 0$.....01 mark

$\overline{AB} = \overline{AE} + \overline{EB}$

$= 13\ cm + 5\ cm$

$= 18\ cm$.....01 mark

(b) $Sin\theta = \frac{12}{13}$, from $Sin^2\theta + Cos^2\theta = 1$

$Cos^2\theta = 1 - Sin^2\theta$

$\sqrt{Cos^2\theta} = \sqrt{1 - Sin^2\theta}$

$Cos\theta = \sqrt{1 - \left(\frac{12}{13}\right)^2}$

$= \sqrt{\frac{169-144}{169}}$

$= \sqrt{\frac{25}{169}}$

(i) Cosθ $= \frac{5}{13}$ .....01 mark

(ii) $Tan\theta = \frac{Sin\theta}{Cos\theta} = \frac{12}{13} \times \frac{13}{5} = \frac{12}{5}$ .....01 mark

10. (a) i. $a^2 - b^2 = [a - b][a + b]$

$17\,972^2 - 17\,872 \times 17\,872 = 17\,972^2 - 17\,872^2$

$= [17\,972 - 17\,872][17\,972 + 17\,872]$……01 mark

$= [100][35\,844]$

$= 3\,584\,400$……01 mark

ii. $S = \pi r\sqrt{r^2 + h^2}$

$\frac{S}{\pi r} = \frac{\pi r}{\pi r}\sqrt{r^2 + h^2}$

$\frac{S}{\pi r} = \sqrt{r^2 + h^2}$

$\left[\frac{S}{\pi r}\right]^2 = \left[\sqrt{r^2 + h^2}\right]^2$

$\frac{S^2}{\pi^2 r^2} = r^2 + h^2$……01 mark

$\frac{S^2}{\pi^2 r^2} - r^2 = r^2 - r^2 + h^2$

$\sqrt{h^2} = \sqrt{\frac{S^2 - r^4\pi^2}{\pi^2 r^2}}$

$h = \pm \frac{\sqrt{[S - \pi r^2][S + \pi r^2]}}{\pi r}$ ……01 mark

(b) $x^2 - 19x - 372 = 0$

Product is -372 and factors are $-31, 12$……$00\frac{1}{2}$ mark

$x^2 - 31x + 12x - 372 = 0$

$x(x - 31) + 12(x - 31) = 0$

$(x + 12)(x - 31) = 0$……$00\frac{1}{2}$ mark

Either $x - 31 = 0$ *or* $x + 12 = 0$

$x = 31$ *or* $x = -12$……01 mark

SECTION B (40 Marks)

11. (a) (i) $5 + 20 + x + 25 + 10 + 5 = 100$

$x = 100 - (5 + 20 + 25 + 15)$

$x = 100 - 65$

$x = 35$……01 mark

Class interval	Frequency	Cumulative frequency	Upper real class boundary
$40 - 42$	5	5	42.5
$43 - 45$	20	25	45.5
$46 - 48$	35	60	48.5
$49 - 51$	25	85	51.5
$52 - 54$	10	95	54.5
$55 - 57$	4	100	57.5
Total	100		

…… ……01 mark

Modal class$= 46 - 48$

Mode $= l + \left(\frac{t_1}{t_1 + t_2}\right) i$

= 45.5 + $\left(\frac{35 - 20}{35 - 20 + 35 - 25}\right)$ 3……$00\frac{1}{2}$ mark

$= 45.5 + \left(\frac{15}{15+10)}\right)3$……$00\frac{1}{2}$ mark

$= 45.5 + \frac{45}{25}$

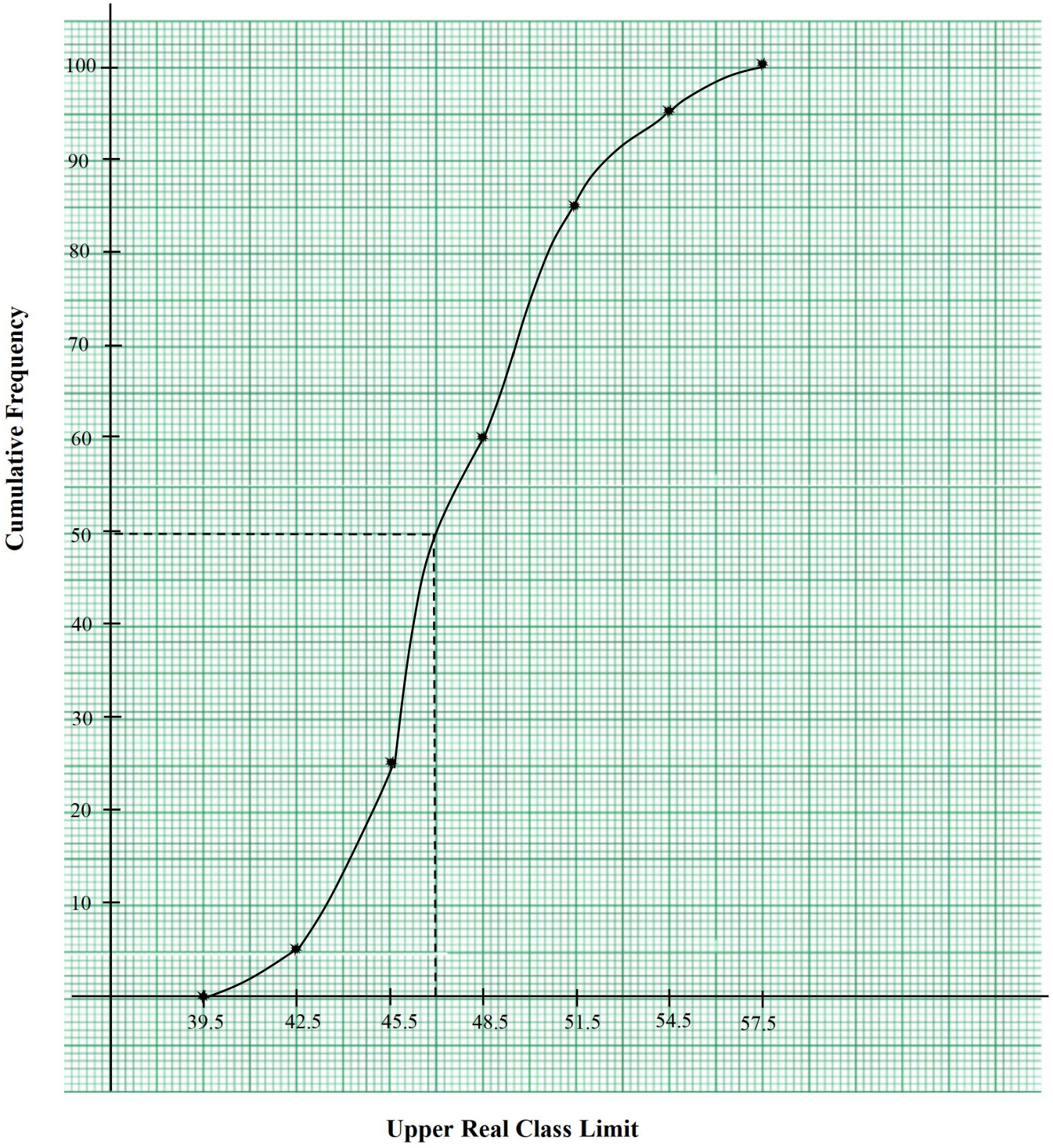
$= 45.5 + 1.8$

$= 47.3$ ……01 mark

(ii) Median position$= \left(\frac{n}{2}\right)^{th} = \left(\frac{100}{2}\right)^{th} = 50\ th$

$= \frac{3}{10} \times 5 = 1.5 = 45.5 + 1.5 = 47$

Therefore estimated median is 47……01 mark



(b) $\overline{CE} \times \overline{ED} = \overline{AE} \times \overline{EB}$ .....00$\frac{1}{2}$mark

$16\ cm \times 6\ cm = 8\ cm \times \overline{ED}$ .....01 mark

$\overline{ED} = \frac{16\ cm \times 6\ cm}{8\ cm}$.....00$\frac{1}{2}$mark

$\overline{ED} = 12\ cm$ .....01 mark

12. (a) i. $x=\frac{-b}{2a}$ ……01 mark.

$=\frac{--5}{2}=\frac{5}{2}$……01 mark.

ii. Minimum value ($a>0$ *i.e* $1>0$)

$y=\frac{4ac-b^2}{4a}$……01 mark.

$=\frac{4\times1\times6-25}{4}=\frac{24-25}{4}=\frac{-1}{4}$……01 mark.

iii. Turning point (x,y)

$=\frac{-b}{2a}, f\left(\frac{-b}{2a}\right)$ *or* $\frac{4ac-b^2}{4a}$……01 mark.

$=\left(\frac{5}{2},\frac{-1}{4}\right)$……01 mark.

iv. $y-$iintercept$=c=6$ ……01 mark.

$x-$iintercepts$=2$ *or* 3 by calculator……01 mark.

$f(x)=x^2-5x+\frac{25}{4}+6-\frac{25}{4}$……01 mark.

$=\left(x-\frac{5}{2}\right)^2+\frac{24-25}{4}$

$=\left(x-\frac{5}{2}\right)^2-\frac{1}{4}$

Therefore $f(x)=x^2-5x+6=\left(x-\frac{5}{2}\right)^2-\frac{1}{4}$……01 mark.

(b) Domain of R$=\{x:x\in\mathbb{R}\}$……02 marks.

Range of R: Make 'x' the subject

$y=3x^2+6$

$y-6=3x^2$

$\frac{3x^2}{3}=\frac{y-6}{3}$

$\sqrt{x^2}=\sqrt{\frac{y-6}{3}}$

$x=\pm\sqrt{\frac{y-6}{3}}$……01 mark.

$=y;\left(\frac{y-6}{3}\right)\geq0$

$=y:3\left(\frac{y-6}{3}\right)\geq0\times3$

$=y;y-6\geq0$

$=y:y-6+6\geq0+6$

$=y:y\geq6$

Therefore range of R=$\{y: y \geq 6\}$……01 mark.

13. (a) $y = x, m = 1\ tan\theta = 1\ tan^{-1}(1) = 45^{\circ}$……$00\frac{1}{2}$ mark

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = M_x = y\begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} cos2\theta & sin2\theta \\ sin2\theta & -cos2\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} cos90^{\circ} & sin90^{\circ} \\ sin90^{\circ} & -cos90^{\circ} \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix}$$……$00\frac{1}{2}$ mark

$$= \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$……$00\frac{1}{2}$ mark

Matrix of rotation for angle θ is

$R_{\theta} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$ Since rotation is 90° clockwise $\theta = -90^{\circ}$

$$R = \begin{bmatrix} \cos(-90^{\circ} & -\sin(90^{\circ} \\ \sin(-90^{\circ} & \cos(-90^{\circ} \end{bmatrix}$$

$R = \begin{bmatrix} cos90^{\circ} & sin90^{\circ} \\ -sin90^{\circ} & cos90^{\circ} \end{bmatrix}$, but $sin(-90^{\circ}) = -sin90^{\circ}$ and $\cos(-90^{\circ}) = \cos(90^{\circ})$

$$= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$……01 mark

Thus the image of point (2,4) under the rotation -90° is

$$\begin{bmatrix} x'' \\ y'' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$……01 mark

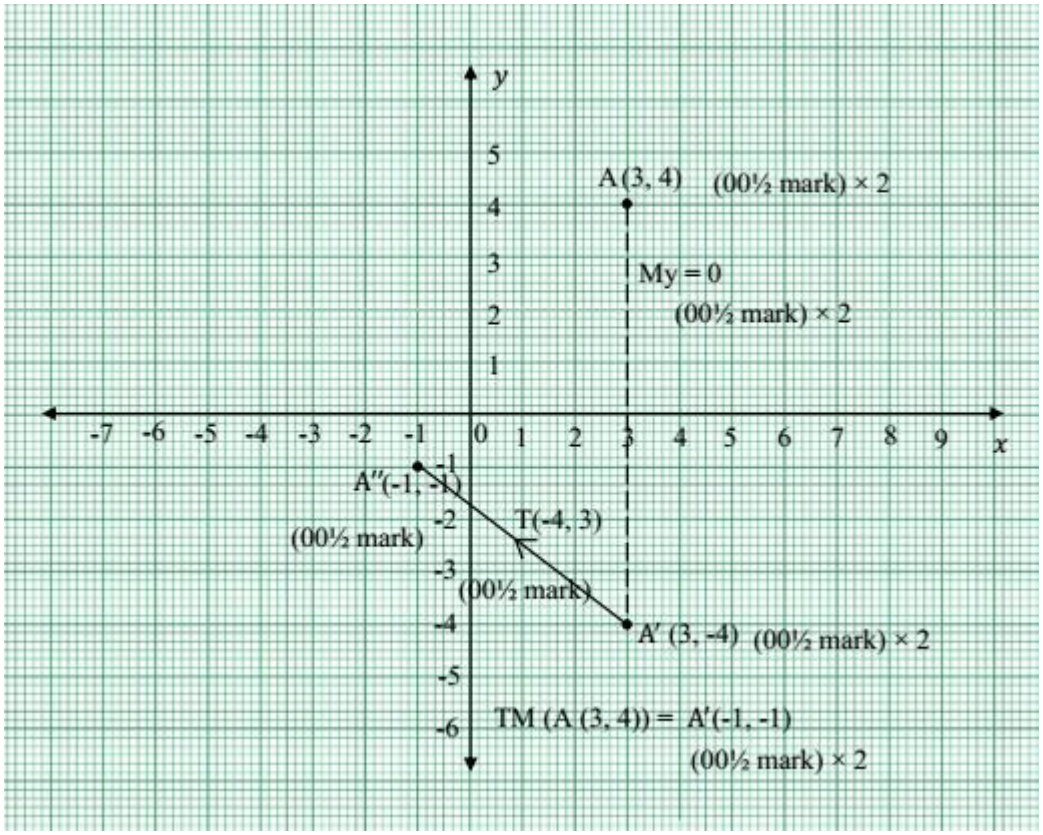
(b) $A(3,4) \xrightarrow{M_{y=0}} A'(3, -4)$ ……01 mark

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} x \\ y \end{pmatrix}$$……01 mark

$$= \begin{pmatrix} -4 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$……01 mark

$$= \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$ …… ..01 mark

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$…… …01 mark



14. (a) $y = x$

x	0	1	2
y	0	1	2

$y = 3$

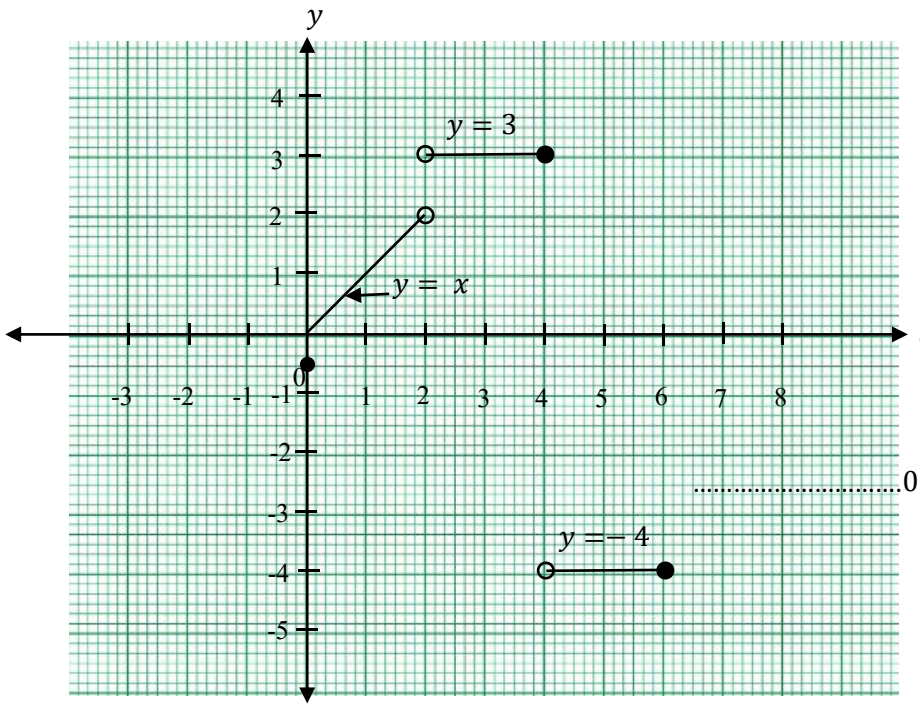
x	2	3	4
y	3	3	3

$y = -4$

x	4	5	6
y	-4	-4	-4

Domain $= \{x: 0 \leq x \leq 6 \ except \ 2 \ or \ 0 \leq x < 2 \ and 2 < x \leq 6\}$……01 mark

Range$= \{y: 0 \leq y < 2, 3, -4\}$……01 mark



.....01$\frac{1}{2}$ marks

(b) $y = x$

x	-2	-1	0	1	2
y	-2	-1	0	1	2

$y = -x$

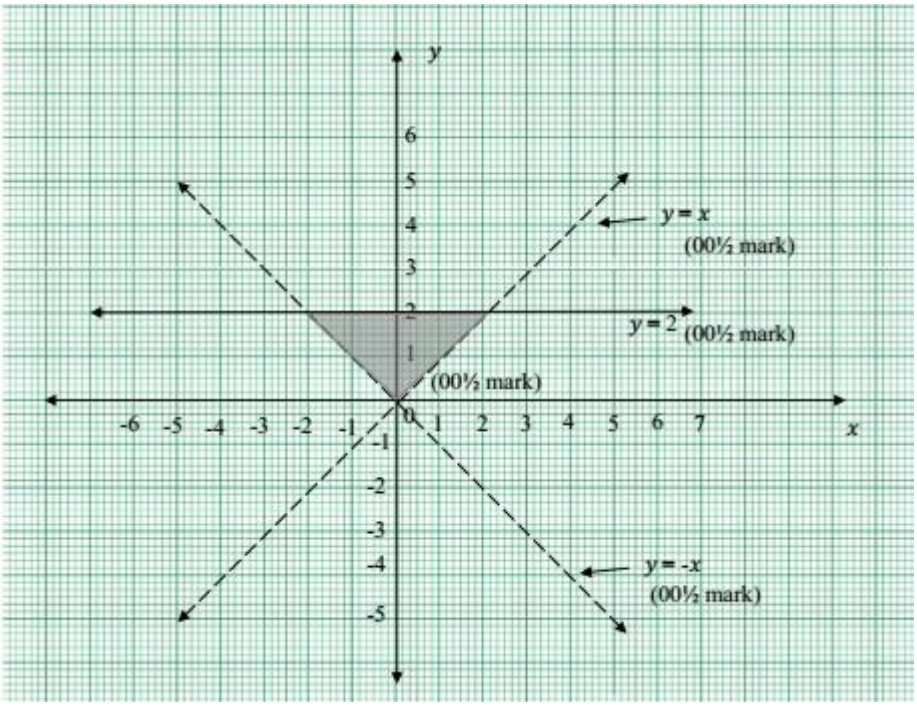
x	-2	-1	0	1	2
y	2	1	0	-1	-2

$y = 2$

x	-2	-1	0	1	2
y	2	2	2	2	2

Domain $= \{x: -2 \leq x \leq 2\}$.....00$\frac{1}{2}$ mark

Range $= \{y: 0 < y \leq 2\}$.....00$\frac{1}{2}$ mark


y
x
y = x
(00½ mark)
y = 2
(00½ mark)
(00½ mark)
y = -x
(00½ mark)